The current lack of certain odd-lot quote, depth of book, and auction information in SIP data could affect price efficiency. The gap in information between data provided by exclusive SIPs and proprietary data products may cause prices in some securities to be less efficient, i.e., to deviate further from fundamental values, if market participants with access to proprietary data products do not incorporate this information into prices quickly enough through their trading or quoting activity. However, the Commission does not know the extent of this possible effect, but it believes the effect could be larger in less actively traded securities where the gap in information between SIP data and proprietary data products is larger.

The Commission believes that, to the extent that there is information in the new core data elements that is not currently reflected in market prices, the final amendments may improve price efficiency.[2443] In particular, the introduction of odd-lot quote, depth of book, and auction information into core data could result in the information becoming impounded in prices more rapidly and accurately as a result of the more widespread dissemination of this information. As the Commission understands that the most sophisticated traders already have access to this information and likely already compete to profit from it, the Commission expects that the size of this gain in price efficiency would be small because this information is already impounded quickly into prices.

Finally, under the current rule, the exclusive SIPs operate like public utilities in their consolidation and distribution of the NMS stock data.[2444] The changes will unbundle the data fees for consolidated market data from the fees for its consolidation and distribution.[2445] The

---

2443 See supra Section V.B.2(a).

2444 See Proposing Release, 85 FR at n.390.

2445 See supra Section V.C.2(c).

decentralized consolidation model will subject the fees charged by competing consolidators for the consolidation and distribution of consolidated market data to competition. The Commission believes that the decentralized consolidation model will lead to consolidated market data being distributed in a timelier, efficient, and cost-effective manner. The Commission believes that the changes to the consolidation and distribution of consolidated data is economically similar to the restructuring of public utilities and may have an impact on the efficiency with which the consolidation and distribution is carried out. In particular, as discussed above, the decentralized consolidation model is anticipated to produce better investment to lower costs and improve quality in the consolidation and distribution of consolidated market data, as well as promote better price competition (all of which translates into a more efficient allocation of capital) than the bidding process currently in place.[2446]

The Commission acknowledges the uncertainty in this conclusion.[2447] The literature on the economics of restructuring public utilities does not provide clear guidance. Some papers show efficiency gains from regulatory restructuring,[2448] yet others claim no efficiency gains or efficiency declines after regulatory restructuring of public utilities.[2449] The likely impact of the adopted changes rests on the strengths and weaknesses of the existing exclusive SIP model.

2446 See id.

2447 One commenter stated that this information "should be essential" to the Commission's analysis. See Nasdaq Letter IV Letter at 47. The Commission continues believe there is uncertainty in its conclusion, but does not believe this precludes the conclusion entirely.

2448 See, e.g., Kira R. Fabrizio et al., Do Markets Reduce Costs? Assessing the Impact of Regulatory Restructuring on US Electric Generation Efficiency, 97 AM. ECON. REV. 1250 (2007).

2449 See, e.g., Severin Borenstein, *The Trouble with Electricity Markets: Understanding California's Restructuring Disaster*, 16 J. ECON. PERSP. 191 (2002).

The Commission believes that the existing exclusive SIP model has an important weakness: it does not provide sufficient competitive incentives.[2450] SIPs have significant market power in the market for core and aggregated market data products and, as a result, do not need to compete to capture demand for their products. The Commission believes that the adoption of the decentralized consolidation model will open up the consolidation and distribution services to data consolidators that will need to vigorously compete to capture some demand for the data they provide. This need to compete for market share will create incentives to reduce costs. As discussed above, the Commission believes that this competition could incentivize competing consolidators to pass on some of those cost savings to customers by charging lower service fees in order to capture market share.[2451] The focus to capture market share might also lead to technological improvements for competing consolidators to be able to differentiate themselves in the eyes of the customers and generate demand.[2452] The Commission believes that these

---

2450 See supra Section V.B.3(a) discussing SIPs market power.

2451 See supra Section V.C.2(b). However, the Commission also acknowledges the possibility that fees for the consolidation and distribution of consolidated market data may remain the same or increase, because consolidated market data will contain more information and/or there might not be enough competition among competing consolidators.

2452 Several studies found evidence of efficiency gains and technological improvements from restructuring in the public utilities sector. In the electricity industry, for example, the introduction of competition to the electricity generation services created strong incentives to become more cost efficient and technologically advanced to improve operating performance. If a plant could not become efficient enough to compete, it would lose business and have to exit the market. Craig and Savage (2013) establish a 9% increase in efficiency in investor-owned electricity plants in response to the restructuring and increasing competition in the electricity sector. Similarly, Davis and Wolfram (2012) argue that electricity market restructuring is associated with a 10 percent increase in operating performance for nuclear plants generating electricity. The authors state that increasing competition led to managers focusing more attention on financial costs of outages. See J. Dean Craig and Scott J. Savage, *Market Restructuring, Competition and*

improvements in data provision technology and the introduction of competitive forces on fees for the consolidation and distribution of consolidated market data could result in a more efficient allocation of capital.

Additionally, the decentralized consolidation model could allow market participants to receive consolidated market data more efficiently. Instead of having to receive separate consolidated market data feeds from two exclusive SIP plan processors, UTP and CTA/CQ Plans, market participants will have the option to receive all of their consolidated market data from one competing consolidator.[2453] This could allow market participants to achieve efficiencies in the design and in making modifications to their systems for the intake of consolidated market data because they will only have to configure their systems to intake consolidated market data from one source.

2. Competition

As discussed previously, the Commission believes this rule will have a substantial impact on competition. The Commission identifies seven markets or areas of the market for which the rule would have a substantial impact on competition. The Commission acknowledges that the seven markets or areas may not be a comprehensive list of all markets or areas for which the rule might have an impact on competition. However, the Commission believes that competition in these seven markets or areas are most likely to be impacted substantially by this rule.

---

the Efficiency of Electricity Generation: Plant-level Evidence from the United States 1996 to 2006, 34 ENERGY J. 1 (2013); Lucas W. Davis and Catherine D. Wolfram, *Deregulation, Consolidation, and Efficiency: Evidence from US Nuclear Power*, AM. ECON. J. APPLIED ECON. 194 (2012).

[2453] The Commission acknowledges that market participants may subscribe to more than one competing consolidator for different core data products or as a backup feed.

First, the adopted rule fosters a competitive environment for the consolidation and dissemination of consolidated market data to replace the centralized consolidation model, which is not currently subject to competitive pressures.[2454] Under the final amendments multiple competing consolidators will be able to distribute consolidated market data products to market participants. The Commission believes that, since market participants could freely select the competing consolidator that charged the lowest distribution fee or offered better quality (i.e., lower latency, a more reliable system), the competing consolidators will be subject to competitive forces and the marketplace for the consolidation and dissemination of consolidated market data products will be competitive if enough competing consolidators enter the market.[2455] As discussed above, the Commission believes that this introduction of competition could reduce the prices competing consolidators charge for the consolidation and distribution of consolidated market data products and improve the quality of consolidated market access.[2456] The Commission recognizes the risk that there could be too few competing consolidators to realize these benefits fully, in which case the adopted competitive changes may have a number of costs,[2457] including higher prices for the consolidation and dissemination of consolidated market data products, which could increase the overall prices market participants pay for consolidated market data.[2458]

2454 See supra Section V.B.3(a).

2455 The Commission assumes that enough competing consolidators will enter the market in order to make it competitive. See supra Section V.C.2(a).

2456 See supra Sections V.C.2(a); V.C.2(b); V.C.2(c).

2457 See supra Sections V.C.2(a); V.C.2(d).

2458 See supra Section V.C.2(a).

One commenter stated that the above characterization of the effects of the amendments on competition represented a "false dichotomy" because the current marketplace already has competition in the form of competing exchanges, and notes that the Commission failed to analyze a comparison with this feature.[2459] This commenter stated that exchanges compete for order flow, and that the sale of proprietary data products is part of this competition, which offers trading services and data in return for the "all-in costs" of trading. This commenter stated that since exchanges compete for order flow, it is incorrect for the Commission to say that there is no competition today, and that there will be competition after the final amendments are implemented. The Commission disagrees that the above characterization is a false dichotomy. The market for the consolidation and dissemination of core data today does not have competition, but rather, exclusive processors in the form of the exclusive SIPs, from which all core data must originate.[2460] Because the final amendments are designed to expand the content and improve the dissemination of core data, the appropriate comparison is to the manner in which core data is processed today, not to the competition between exchanges for trading services.

The Commission recognizes that Rule 614(d)(9) and the addition of the definition of SCI competing consolidator to Regulation SCI could impact competitive dynamics in the competing consolidator market. If the exclusive SIPs become competing consolidators, they could gain a competitive advantage over other competing consolidators with respect to Regulation SCI

2459 See Nasdaq Letter IV at 48. See also Nasdaq Letter IV at 32, describing proprietary data products as competitive.

2460 See supra Section V.B.3(a) (discussing the exclusive nature of the SIP processors).

compliance because they would face lower barriers to entry since they are currently SCI entities and already incur many of these costs.[2461] Comparatively, the Commission believes the costs associated with Rule 614(d)(9), along with the costs associated with later potentially being an SCI competing consolidator, could raise the barriers to entry for firms seeking to become competing consolidators who are not already exclusive SIPs.[2462] Therefore, Rule 614(d)(9) and the addition of the definition of SCI competing consolidator to Regulation SCI could result in fewer firms seeking to become competing consolidators, which could lead to less competition in the competing consolidator market. Less competition and less innovation would reduce the incentives of competing consolidators to reduce the costs and improve the speed and quality of their consolidated market data aggregation and dissemination services. Additionally, after the initial transition period, the Commission believes that the lower burdens associated with Rule 614(d)(9) could potentially give a competing consolidator below the SCI competing consolidator threshold a competitive advantage over SCI competing consolidators because it would have lower compliance costs. However, the Commission does not believe that this competitive advantage will be significant because a competing consolidator with market share below the threshold that gained market share would become an SCI competing consolidator after it crossed the threshold.

2461 See supra Section V.C.2(e)(ii) for a discussion of costs related to Regulation SCI and Rule 614(d)(9). See supra Section V.C.2(a)(i)a for additional discussion of other factors affecting the barriers to entry for competing consolidators.

2462 SROs that do not operate an exclusive SIP could also be at a competitive disadvantage relative to an SRO that operates an exclusive SIP that became a competing consolidator, because they would face higher initial SCI related costs than an exclusive SIP would if it became a competing consolidator. See supra Section VI.C.2(a)(i)b.; supra Section V.C.2(e)(ii).

The limited exemptive relief from the rule filing requirements of Section 19(b) of the Exchange Act and Rule 19b-4 thereunder, Section 19(d), the requirements of Section 6(b), and from Regulation SCI provided to competing consolidators affiliated with exchanges will reduce the regulatory burdens that would otherwise be faced by such an entity in becoming a competing consolidator.[2463] As a result, SROs and their affiliates may find it less burdensome to operate a competing consolidator, and therefore may be more likely to enter the market, which will promote competition in the provision of consolidated market data.

Additionally, the Commission believes that the public disclosure of the information in Form CC and the performance metrics and operational information competing consolidators will provide on their websites would enhance competition between competing consolidators.[2464] The public disclosure of competing consolidator fees and performance metrics will allow market participants to more easily compare competing consolidators and select the ones that charged the lowest fees or offered the best performance. This could enhance competition between competing consolidators. For example, if the public disclosures show that certain competing consolidators have higher fees or poor performance, it may result in those competing consolidators losing subscribers and earning lower revenues. Similarly, competing consolidators who display lower prices or superior system performance may be able to attract more subscribers and earn more revenue. This, in turn, could enhance competition by incentivizing competing consolidators to lower fees and/or innovate and make investments in their systems in order to improve system performance in order to attract more subscribers. The Commission acknowledges that the public

---

2463 The Commission believes that competing consolidators affiliated with exchanges will choose to operate under the provisions of the exemption. See supra Section V.C.2(a)(i)a for additional discussion of the impact of the limited exemptive relief on barriers to entry.

2464 See supra Section V.C.3.

disclosure of Form CC could harm competition by making firms reluctant to enter the competing consolidator market and reducing the incentives of competing consolidators to innovate if it discloses certain information that a competing consolidator might view as a "trade secret" or giving it a competitive advantage. However, the Commission believes that these effects are not likely to occur because the disclosures on Form CC are not detailed enough to allow other market participants to reproduce a competing consolidator's "trade secret."[2465] Additionally, the delayed public disclosure of material amendments to Form CC should prevent another competing consolidator from replicating a competing consolidator's innovations before it has a chance to implement them.[2466]

The Commission recognizes that the registration process for Form CC could create uncertainty about whether a Form CC would be declared ineffective. This could potentially harm competition in the market for competing consolidators by raising the barriers to entry and creating a disincentive for entities to become competing consolidators. However, the Commission believes that these effects will not be significant because the Commission will not declare a Form CC ineffective without notice and opportunity for hearing. Additionally, entities whose Form CC is declared ineffective will still have the opportunity to file a new Form CC with the Commission.

The Commission considered the effect of the interaction between the proposal and the CAT NMS Plan on competition among competing consolidators, but believes that this interaction will not have a significant effect on the competitive landscape. In particular, the Commission considered two effects: first, the effect in the event that there is a bias toward an

---

[2465] See supra Section III.C.8(e)(ii) for additional discussion about Form CC disclosures.

[2466] See supra Sections III.C.7, V.C.3.

exchange-operated competing consolidator over other competing consolidators and second, any competitive advantage for the competing consolidator selected for the CAT NMS Plan. In relation to any bias, the Commission notes that the CAT NMS Plan will be only one of many potential customers of the competing consolidator, so this bias is not likely to affect the market unless the selection produces a competitive advantage. In particular, a competing consolidator could enjoy a competitive advantage only if broker-dealers believe that market surveillances would be less likely to appear to show violations if the broker-dealers made trading decisions using the same data used in SRO surveillances. However, the latency differences across the competing consolidators are likely to measure in the microseconds while the clock synchronization requirements for industry members in the CAT NMS Plan is 50 milliseconds for electronic order flow.[2467] Therefore, the Commission does not believe the CAT's choice of competing consolidator will confer any regulatory value on the competing consolidator or their broker dealer clients.

Second, the Commission believes that the expanded content and reduced latency of consolidated market data will make it a more viable substitute for proprietary data feeds.[2468] The Commission believes that this will increase competition between consolidated market data and exchange proprietary data feeds. These competitive pressures could lead to lower prices for proprietary data feeds and may reduce the data costs that market participants pay, at the expense

---

2467 See CAT NMS Plan, supra note 1220, at Section 6.8.

2468 However, consolidated market data would not be a perfect substitute for the proprietary data feeds because it would not contain all the information in proprietary data feeds. For example, the expanded core data would not include full depth of book information or information on all odd-lots. See supra Section V.C.4.